Dragan Stosic, MSc

Senior Software Engineer | Formal Verification & AI Systems

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Summary

Senior software engineer combining production AI systems with machine-checked verification in Lean 4 and PVS. Contributor to NASA's PVS Library and the DARPA-funded BESSPIN voting-system demonstrator. Develops formal models of certification, repair and probabilistic algorithms, alongside LLM workflow compilers and validated incremental editors. Recent work combines Lean proofs of a functional model with CBMC verification of a bounded sequential C API.

Core Skills

Formal verification: Lean 4, PVS, CBMC; refinement, invariants, probabilistic algorithms, proof replay and axiom-dependency audits. Earlier work with Coq, Z3, Frama-C, JML and VHDL.

AI and software: Python, C, C#, Bash; LLM orchestration (AWS Bedrock, Claude, GPT), structured outputs, workflow compilation, incremental editing, pytest and Hypothesis.

Platforms: AWS, Databricks/Spark, Delta Lake, PostgreSQL/pgvector, Redis, OpenSearch, Docker, Terraform and CI/CD.

Selected Formal Verification Work

Lean 4 / PVS: Certification, Refinement and Repair

Independent research | 2024–2026

- Developed a Lean 4.35.0 library with **1,824 explicit theorems and lemmas across 191 source modules**, reproducible builds and audited axiom dependencies; no project-owned Lean axioms or `sorry` dependencies.
- Developed independently replayed PVS counterparts for all 1,824 source results. Topics include certificate-preserving transitions, parametric certification, repair and progress obligations, the General Lovász Local Lemma and Moser–Tardos constructive repair.
- Made preservation assumptions explicit: admitted transitions preserve selected certificates; repair and probabilistic progress require separate proof obligations.

Lean source and checks | PVS source and proof replay

C Buffer Publication: Lean Model and CBMC Contracts

Verification case study | 2026

- Verified **eight contracts** of a sequential C buffer API with a 64-byte capacity using CBMC, with **seven negative controls** to check that intended violations are detected.
- Proved the functional model and certificate-preserving controller refinement in Lean. C verification covers the API under a declared C/library model; compiler correctness and concurrency are outside its scope.

Source and reproducible verification reports available on request.

NASA PVS Library: External Contributor

2013

- Contributed refinement-relation definitions and lemmas in `sets_aux/rr_rel.pvs`; named author in the source and listed contributor to NASA's PVS Library.

NASA contributor list | rr_rel.pvs

BESSPIN Voting System Demonstrator: External Contributor

With Galois Inc. / DARPA-funded project | 2019–2020 | Remote

- Contributed formal verification and hardware/software attack-surface analysis to the open-source election-security demonstrator. Tools included PVS, Coq and Frama-C.

Project and contributor credits

Industry Experience

Rival (ex-SilkRoad Technology): Senior AI Engineer

2021–2026 | Remote, US company

Built enterprise HR automation spanning workflow generation, conversational editing, retrieval and production ML.

- Designed LLM workflow generation from natural-language requests and Excel, with modular skill routing and a **deterministic topology compiler** that preserves supplied dependencies and validates references, sequencing and phase gates.
- Built a keyed-delta editor with identity mapping, dependency validation, version checks, request deduplication, tenant-scoped sessions and **structural undo / rollback**.
- Created pytest and Hypothesis suites for identity bijection, round-trip preservation, deterministic output, dependency constraints, duplicate prevention and undo/replay.
- Delivered a RAG assistant over **3,400+ documents** using Bedrock and pgvector/HNSW; implemented Garak testing, input moderation, PII anonymization and Bedrock Guardrails.
- Owned a **600+ notebook Databricks platform**, Delta Lake processing and resumable Delta-to-OpenSearch publishing with pinned snapshots, retry records and load control.
- Trained and served an XGBoost response-likelihood model and attention-based TensorFlow/Keras BiLSTM career-trajectory models across eight job families.

IBM Ireland: Senior Engineer

2011–2019 | Dublin

Watson IoT and industrial analytics: Spark ML pipelines, Random Forests and Logistic Regression over Hadoop/Hive, with Flask APIs.

University College Dublin: Researcher

2009–2011 | Dublin

Formal specification with JML and PVS, VHDL hardware verification and the SenseTile sensor-platform project.

Earlier experience: ProCredit Bank (Java/C# banking), FinSoft London (European Central Bank portal, J2EE), Institute Mihailo Pupin (real-time embedded systems, DSP and VHDL).

Selected Research Outputs

- **Empirical Information Geometry on an Embedded Adaptive Lighting System: Multi-Chart Manifold Transitions Under Signal Collapse.** Zenodo research report, 2026. DOI: 10.5281/zenodo.20094759
- **Excitation-Dependent Observability Geometry on an Embedded Adaptive Lighting Manifold.** Zenodo preprint, 2026. DOI: 10.5281/zenodo.20389804
- **Optimal Transport Geometry of Natural Spectral Regime Transitions.** Zenodo preprint, 2026. DOI: 10.5281/zenodo.21956336

Education and Availability

MSc, Faculty of Electrical Engineering, University of Belgrade (1994).

Available immediately for remote B2B work through Trustworthy Systems d.o.o. or an EOR. Belgrade / Central European time; working-hours overlap with US East Coast teams.